



Types of Memory Transcript

Hello everyone, welcome to BITS PLANET DIGITAL. We will be continuing our learning journey of data visualisation and storytelling. Today we will be looking at principles of effective visual design and continue to learn further concepts.

We will be looking at the types of memory and the pre-attentive attributes and their usage in text and graphs. Remember when we said your data visualisations have to be audience centric which means you have to place audience at the centre of your data visualisations while designing them and see how they will pursue your data visualisations. In the same spirit we will try to understand what is the role of memory that audience possess and how it interacts with the visualisation process.

What we mean by this and we are saying the visualisations have to be data sent audience centric not data centric. We also have to account the memory capacity of audience and how it interacts with the perception process and best design our visuals so that it assimilates with the optimal memory utilisation process for the audience. In this context we will try to understand what are the different types of memories that the audience use for processing information presented in our visuals.

Generally the human memory is categorised into three types iconic memory, short term memory and long term memory. Iconic memory is that part of the memory which is used for very short type but has very large capacity as opposed to the short and long term memories which are used on a different time scale relatively for a longer period compared to iconic memory. You can say iconic memory is the first representation you can say iconic memory is the first thing that the humans use or the audience use to pursue your visual.

It is very brief but very fast it perceives visuals and starking colour or pre-attentive attributes and therefore it quickly grabs the content as opposed to the other two forms of memory short term and long term which take little longer to process the information. Now let us dwell deep into the characteristics of these three types of memories. We have iconic memory which is used for a fraction of second which is typically for a fraction of second then short term memory you have which is for few seconds and long term memory which can be for a few minutes to lifetime.

This is the span of memory if you see iconic memory is very short but very effective short term memory is also brief long term memory is in fact very long which you can remember for a long time. However iconic memory is focused on the visuals and the pre-attentive attributes. Then short term memory also processes visual chance but small amounts of data can be stored in short term memory.

Long term memory has both visual and verbal content. The capacity of iconic memory is very large but it is very finite in the sense it is very very brief as opposed to short term memory which holds small relatively small amounts of information long term

memory can withhold unlimited information. The processing of iconic memory is unconscious it does not happen in the conscious part of the brain whereas short term memory and long term memory are both conscious and mix of conscious and unconscious.

The purpose of iconic memory is to quickly detect changes in the environment whereas the short term memory and long term memory have different purposes like temporary holding and storage of patterns and experience. The iconic memory you have to use strong visuals and contrasts to appeal to iconic memory short term memory and long term memory require different approaches for clear attending to short term memory to get best out of short term memory you have to avoid clutter. So that the information observed is stored properly the small amount of information is readily and correctly captured whereas long term memory you can use visual and verbal recall to reinforce the learnings.

By understanding the type of memory that your audience possess you can reconsider your visual design practises or the visualisations to best deliver the message. Let us say for example the attention is to create immediate action I can give you the example of the traffic lights when you see a red traffic light you immediately stop that is iconic memory immediately reacting to the same thing can be achieved by writing a stop sign which would definitely take more time to react. It is the same thing that is play in this visualisation space also when we are appealing to the iconic memory we are using strong visuals cues and signals to drive home a point very quickly whereas in the other two types of memory there is some amount of processing lag that we accept and try to create our visualisations.

Now let us see how to use the pre attentive attributes to effectively design our visuals again pre attentive attributes are used by the iconic memory to create impact. Now look at the example on the slide these are the numbers presented in one form and the task is to count the number of digit threes that are occurring in this particular diagram. If I were to ask you to do this it would definitely take some time and cognitive load to count the number of threes.

Now look at this diagram as opposed to this where the threes are highlighted if I ask you to count the number of threes in diagram definitely it would take much lesser time than in the first case. Here we have used highlight as a pre attentive text to draw your attention to counting the number three which is the task. So simple highlighting of text or the content brings more attention to the detail or the visual therefore enabling a faster execution of the task.

Cognitively we see things that are highlighted than the things which are similar. So if you want to drive attention you can simply highlight and bring the attention of your audience to that particular point. This is a simple example of pre-attentive attribute in this case the highlighting of the text.



Now we will look at different types of pre-attentive attributes and how they can be assimilated into your visuals. For example orientation all the lines are straight one line is slanted you directly would notice the deviation. You can also refer to the gestalt principle of focus point where everything similar and a point that is different or focused will get the immediate attention.

You can also change the shape among the continuum and give a different effect. You can also change the length of the line and create an impact. Similarly with thickness different thickness can give a different highlighting or meaning.

For example when you have multiple series you can use thickness of the line to reflect the importance of the series as compared to others. If you remember we have seen this in the example of campaign fundraising where the current campaign fundraising line was thicker compared to the last year's line. Also we can use the size differentiation, curvature, similarly added marks, enclosure which is also an important pre-attentive attribute where you create an enclosure rounding an object to get attention of the audience to that particular space.

You can also use hue which is a different colour than the other colours and ascertain a meaning to that colour. Again I would bring you to the gestalt principles of similarity. Human brain naturally categorises objects which are in similar form and colour as belonging to one particular group.

Let us say the combination of a scatter plot you have used hue and coloured the dots into two or three different colours. Naturally the brain would categorise them as belonging to three different groups and this is use of pre-attentive attribute in scatter plot to impose categories in the same plot. Likewise you can use intensity the differentiation between dark and light colours.

You can also use the spatial position to distinguish and finally motion to show the pre-attentive attributes which in animation can be used. Now again a word of caution we should not overdo the pre-attentive attributes. We should use them in an assimilative fashion to the design so that the final visual is more aesthetically pleasing and gives the more clarity to the design or visual that we are trying to build.

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